

Simple Linear Regression Manual Calculation

Part 2

Simple 5-Point Manual Calculations

Expanding to 5 rows reveals the true power of Linear Regression. Unlike a 2-point dataset which draws a perfect line between two dots, a 5-point dataset introduces "noise." The algorithm must now find the mathematical center of gravity—a line of best fit that minimizes the overall distance (residuals) to all five points simultaneously.

The Dataset (Engine Size vs. CO2)

Observation	X (Engine Size)	Y (CO2 Emissions)
Row 1	1	110
Row 2	2	140
Row 3	3	200
Row 4	4	260
Row 5	5	290

Step-by-Step Architectural Breakdown

Step 1: Calculate the Means ($\bar{x}$ and $\bar{y}$)

How: Add all 5 X values and divide by 5. Do the same for Y.

Why & What: This establishes the 'center of gravity' for your dataset. The best-fit line **MUST** pass exactly through this coordinate point ($\bar{x}$, $\bar{y}$).

[Workspace for calculation]

Step 2: Calculate Deviations ($X - \bar{x}$) and ($Y - \bar{y}$)

How: For each of the 5 rows, subtract the mean of X from the row's X value, and the mean of Y from the row's Y value.

Why & What: This shifts our perspective to look at 'how much bigger or smaller is this engine compared to the average?'

[Workspace for calculation]

Step 3: Calculate the Variance of X: $\Sigma(X - \bar{x})^2$

How: Square the X deviation for each row, then add all 5 squared numbers together.

Why & What: Variance measures the total 'spread' of the X data, penalizing points far from the center. This forms the denominator of our slope.

[Workspace for calculation]

Step 4: Calculate the Covariance of X and Y: $\Sigma[(X - \bar{x})(Y - \bar{y})]$

How: For each row, multiply its X deviation by its Y deviation. Then, add all 5 results together.

Why & What: Covariance proves the relationship. This forms the numerator of our slope, capturing the synchronized movement between the two columns.

[Workspace for calculation]

Step 5: Calculate the Slope ($m = \text{Covariance} / \text{Variance}$)

How: Divide the total from Step 4 by the total from Step 3.

Why & What: This tells us the exact rate of change: 'For every 1 unit increase in Engine Size, how much do Emissions increase on average?'

[Workspace for calculation]

Step 6: Calculate the Intercept ($c = \bar{y} - m * \bar{x}$)

How: Multiply the slope (m) by the mean of X ($\bar{x}$), and subtract that from the mean of Y ($\bar{y}$).

Why & What: This anchors the line to the grid, guaranteeing it passes perfectly through the dataset's center of gravity.

[Workspace for calculation]

Answer Key & Mathematical Verification

Step 1 (Means):

$$X \text{ sum} = 15 \rightarrow \bar{x} = 3$$

$$Y \text{ sum} = 1000 \rightarrow \bar{y} = 200$$

Step 2 (Deviations):

$$\text{Row 1: } X_{\text{dev}} = -2 \mid Y_{\text{dev}} = -90$$

$$\text{Row 2: } X_{\text{dev}} = -1 \mid Y_{\text{dev}} = -60$$

$$\text{Row 3: } X_{\text{dev}} = 0 \mid Y_{\text{dev}} = 0$$

$$\text{Row 4: } X_{\text{dev}} = 1 \mid Y_{\text{dev}} = 60$$

$$\text{Row 5: } X_{\text{dev}} = 2 \mid Y_{\text{dev}} = 90$$

Step 3 (Variance of X):

$$4 + 1 + 0 + 1 + 4 = 10$$

Step 4 (Covariance of XY):

$$180 + 60 + 0 + 60 + 180 = 480$$

Step 5 (Slope m):

$$480 / 10 = 48$$

Step 6 (Intercept c):

$$200 - (48 * 3) = 56$$

Step 7 (Final Equation):

$$y = 48x + 56$$

Verification check:

Unlike the 2-row dataset, this line does not touch every point perfectly. Let's test Row 1 ($X=1$):

Predicted $Y = 48(1) + 56 = 104$.

Actual $Y = 110$.

The model missed by exactly 6 units (the residual). The algorithm specifically chose $m=48$ and $c=56$ because this combination mathematically minimizes the squared residuals across all 5 rows combined.